BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics (Objective Type) — Inter-(Part-II)-A-2021 (For all sessions) — Paper Code: 8191

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8191

Note: NOTE: Write answers to the questions on objective answer sheet provided. Four possible answers A,B,C & D to each question are given. Which answer you consider correct, fill the corresponding circle A,B,C or D given in front of each question with Marker or pen ink on the answer sheet provided.

Q1. If $g(x) = 1/x^2$, $x \neq 0$ then gog(x) equals.

(A) x
(B) x^2
(C) x^4
(D) x^3

Q2. $\lim(\theta \to 0) (1-\cos\theta)/\theta$ equals.

(A) zero
(B) 1
(C) 2
(D) 3

Q3. The derivative of $\sqrt{x}$ at x = 1 is:

(A) 1/2
(B) 2
(C) 1
(D) -1/2

Q4. d/dx[1/g(x)] equals.

(A) $1/g^2(x)$
(B) $-g'(x)/(g(x))^2$
(C) -g(x)
(D) 1/g(x)

Q5. If $y = 5e^x$ then y_3 equals.

(A) $25e^x$
(B) $75e^x$
(C) $15e^x$
(D) $5e^x$

Q6. If f(x+h) = cos(x+h) then f'(x) equals.

(A) cos x
(B) -cos x
(C) -sin x
(D) sin x

Q7. Inverse of ∫.....dx is:

(A) d/dy
(B) d/dx
(C) dy/dx
(D) dx/dy

Q8. $\int_a^b f(x)\,dx$ equals:

(A) $-\int_{\beta a} f(x)\,dx$
(B) $\int_{-\beta a} f(x)\,dx$
(C) $\int_{\beta}^{-a} f(x)\,dx$
(D) $\int_{a}^{-\beta} f(x)\,dx$

Q9. The general solution of dy/dx = -y/x is:

(A) xy = c
(B) $x^2y^2 = c$
(C) x/y = c
(D) y/x = c

Q10. $\int e^{-x}(\cos x - \sin x)dx$ equals:

(A) $-e^{-x} \sin x + c$
(B) $e^{-x} \cos x + c$
(C) $e^{-x} + c$
(D) $e^{-x} \sin x + c$

Q11. The distance of point (3,7) from x-axis is:

(A) 3
(B) 7
(C) -3
(D) -7

Q12. Slope of Y-axis is:

(A) zero
(B) 1
(C) 2
(D) undefined

Q13. Equation of horizontal line through (7,-9) is:

(A) y = -9
(B) y = 7
(C) x = -9
(D) x = 7

Q14. (0,2) is solution of inequality.

(A) 3x+5y > 7
(B) 3x+5y < 7
(C) x < 0
(D) x > 0

Q15. Centre of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is:

(A) (g,f)
(B) (-g,f)
(C) (0,0)
(D) (-g,-f)

Q16. Equation of Latus rectum of parabola $x^2 = 4ay$ is:

(A) $y = -a$ (B) $y = a$

(C) $x = -a$ (D) $x = a$

Q17. Vertices of $x^2/16 - y^2/25 = 1$ are:

(A) $(0,\pm4)$ (B) $(0,\pm5)$

(C) $(\pm4,0)$ (D) $(\pm5,0)$

Q18. The non zero vectors $\underline{a}$ and $\underline{b}$ are parallel if $\underline{a}\times\underline{b}$ is:

(A) zero (B) 1

(C) 2 (D) 3

Q19. $\cos\theta$ equals:

(A) $\underline{a}.\underline{b}$ (B) $\underline{a}\times\underline{b}$

(C) $|\underline{a}\times\underline{b}|$ (D) $\hat{a}.\hat{b}$

Q20. If any two vectors of scalar triple product are equal then its value is:

(A) -1 (B) zero

(C) 1 (D) 2

Mathematics (Essay Type) — Inter-(Part-II)-A-2021 (For all sessions)

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Q. Section -I

Q2. Write short answers of any eight parts from the following.

2x8=16

(i) If $f(x) = x^2 - x$, find (a). $f(-2)$ (b). $f(x-1)$

(ii) Find $\lim(x\to2)\ (x^3-8)/(x^2+x-6)$

(iii) Find $\lim(\theta\to0)\ (1-\cos\theta)/\sin\theta$

(iv) Differentiate w.r.t "x": $(\sqrt{x} - 1/\sqrt{x})^2$

(v) Find dy/dx if $3x + 4y + 7 = 0$

(vi) Differentiate w.r.t "x" $\cos\sqrt{x} + \sqrt{(\sin x)}$

(vii) Differentiate w.r.t "x" $\cot^{-1}(x/a)$.

(viii) If $y = \log_{10}(ax^2+bx+c)$, then find dy/dx

(ix) If $y = x^2.e^{-x}$, then find d^2y/dx^2.

(x) Apply Maclaurin series, Prove that $\cos x = 1 - x^2/2! + x^4/4! + _____ \infty$

(xi) If $f(x) = \sqrt{(x+1)}$ and $g(x) = 1/x^2$, then find (a). (fog)(x) (b). (gof)(x)

(xii) Find the intervals in which f(x) is increasing or decreasing $f(x) = \cos x$, $x \in (-\pi/2, \pi/2)$.

Q3. Write short answers of any eight parts from the following.

2x8=16

(i) Using differential find dy/dx, if $x^2 + 2y^2 = 16$

(ii) Evaluate $\int x\sqrt{(x^2-1)}\, dx$

(iii) Evaluate $\int (1-\sqrt{x})^2/\sqrt{x}\, dx$

(iv) Evaluate $\int \sin^2 x\, dx$

(v) Evaluate $\int (ax+b)/(ax^2+2bx+c)\, dx$

(vi) Evaluate $\int e^{3x}((3\sin x - \cos x)/\sin^2 x)\, dx$

(vii) Sove [sic] $dy/dx = (y^2+1)/e^x$

(viii) Find an equation of the vertical line through (-5,3).

(ix) Find an equation of the line through (-5,-3), (9,-1)

(x) Convert $4x + 7y - 2 = 0$ in normal form.

(xi) Find the area below the curve $y = 3\sqrt{x}$ and above the x-axis between x = 1 and x = 4.

(xii) Find the mid point of the line segment joining the points A(3,1), B(-2,-4).

Q4. Write short answers of any nine parts from the following.

2x9=18

(i) Graph the solution set by shading of inequality $5x - 4y \leq 20$.

(ii) Find equation of circle with centre at $(\sqrt{2}, -3\sqrt{3})$ and radius $2\sqrt{2}$.

(iii) Write equation of tangent to the circle $3x^2 + 3y^2 + 5x - 13y + 2 = 0$ at (1, 10/3).

(iv) Find vertex of $x^2 - 4x - 8y + 4 = 0$.

(v) Find point of intersection of conics $3x^2 - 4y^2 = 12$ and $3y^2 - 2x^2 = 7$.

(vi) Find equation of parabola whose focus is F(-3,4) and directrix is $3x - 4y + 5 = 0$.

(vii) Find the unit vector in the same direction of vector V = [3, -4].

(viii) If $\bar{AB} = \bar{CD}$ find the co-ordinate of the point A when points B, C, D are (1,2) (-2,5) and (4,11) respectively

(ix) Find $|3\underline{v} + \underline{w}|$ if $\underline{u} = i + 2j - k$, $\underline{v} = 3i - 2j + 2k$, $\underline{w} = 5i - j + 3k$.

(x) Find a vector of length 5 in the direction opposite that of $\underline{v} = i - 2j + 3k$.

(xi) Compute $\underline{b}\times\underline{a}$ if $\underline{b} = i - j + k$, $\underline{a} = 2i + j - k$.

(xii) Find the work done if the point at which the constant force $\bar{F} = 4i + 3j + 5k$ is applied to an object, moves from $p_1(3,1,-2)$ to $p_2(2,4,6)$.

(xiii) If $\underline{a} + \underline{b} + \underline{c} = 0$ then prove that $\underline{a}\times\underline{b} = \underline{b}\times\underline{c} = \underline{c}\times\underline{a}$.

Q. Section -II — 10x3=30

Note: Attempt any three questions from the following.

Q5. (a) If $f(x) = \{3x-1 \text{ if } x<1;\ 4 \text{ if } x=1;\ 2x \text{ if } x>1\}$, then show f(x) is continuous at x=1.

(b) If $x = a(1-t^2)/(1+t^2)$, $y = 2bt/(1+t^2)$, then find dy/dx.

Q6. (a) Find the approximate increase in the volume of a cube of the length of its each edge changes from 5 to 5.02.

(b) Determine the value of P such that the lines $2x - 3y - 1 = 0$, $3x - y - 5 = 0$ and $3x + py + 8 = 0$ meet at a point.

Q7. (a) Evaluate $\int_2^3 (x - 1/x)^2\, dx$.

(b) Minimize $z = 2x + y$ subject to the constraints $x + y \geq 3$, $7x + 5y \leq 35$, $x \geq 0$, $y \geq 0$.

Q8. (a) Write equations of two tangents from (2,3) to the circle $x^2 + y^2 = 9$.

(b) Prove by vector method $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$.

Q9. (a) Show that $\cos(x+h) = \cos x - h\sin x - h^2/2!\cos x + h^3/3!\sin x + ...$

(b) Show that an equatin [sic] of the parabola with focus at $(a\cos\alpha, a\sin\alpha)$ and directrix $x\cos\alpha + y\sin\alpha + a = 0$ is $(x\sin\alpha - y\cos\alpha)^2 = 4a(x\cos\alpha + y\sin\alpha)$.